CS432: Databases

Introduction to Databases

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4

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Database Languages

- **Data-Definition Language**
  - To describe the schema
  - DDL compiler generates a set of table templates stored in a data dictionary
  - Data dictionary contains metadata (i.e., data about data)
    - Database schema
    - Integrity constraints
    - Primary key (ID uniquely identifies instructors)
    - Authorization (Who can access what)

Example

```
create table instructor (
        ID char(5),
        name varchar(20),
        dept_name varchar(20),
        salary numeric(8,2))
```

Database Languages

- **Data-Manipulation Language**
  - It enables users to access or manipulate data (retrieve, insert, delete, and modify)
  - DML that involves information retrieval is called a **query language**.
    - **Procedural DMLs:** What data are needed and how to get those data.
    - **Declarative DMLs:** What data are needed without specifying how to get those data.

SQL

- One of the popular commercial language
- Not as powerful as a universal Turing machine
- It does not support:
  - Input from users
  - Output to displays
  - Communication over the network
- Above actions must be written in a host language, such as C, C++, C#, Java or Python etc.

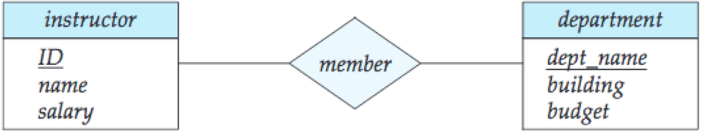
Database Design

It mainly involves the design of the database schema.

- Organizational requirements
- Conceptual schema
- Final design process
  - Logical-design phase
  - Physical-design phase
- Relational model - conceptual design process: (What) attribute want to capture in database and **(How)** to group these attribute to form the various table ("how" part is computer science problem).

Possible designing ideas (for how part)

- Entity-Relationship Model



Possible designing ideas (for how part)

- Normalization

ID	*name*	*salary*	*dept_name*	*building*	*budget*
22222	Einstein	95000	Physics	Watson	70000
12121	Wu	90000	Finance	Painter	120000
32343	El Said	60000	History	Painter	50000
45565	Katz	75000	Comp. Sci.	Taylor	100000
98345	Kim	80000	Elec. Eng.	Taylor	85000
76766	Crick	72000	Biology	Watson	90000
10101	Srinivasan	65000	Comp. Sci.	Taylor	100000
58583	Califieri	62000	History	Painter	50000
83821	Brandt	92000	Comp. Sci.	Taylor	100000
15151	Mozart	40000	Music	Packard	80000
33456	Gold	87000	Physics	Watson	70000
76543	Singh	80000	Finance	Painter	120000

- Repetition of information
- Inability to represent certain information

Database Engine

Partitioned into **three modules** to deal with the responsibilities of the overall system.

- **Storage manager**:
  - Huge size of databases from GBs to several TBs of data.
  - Keeping everything in RAM not possible.
- **Query processor**
  - helps the database system to simplify and facilitate access to data
  - translate queries in non-procedural language into an efficient sequence of operations.
- **Transaction manager**
  - What if database fails?
  - How can it supports multiple users concurrently?

Storage manager

An **interface** between low-level data stored in the database and the application programs and queries submitted to the system.

- Interacts with the file system provided by the OS.
- Storing, retrieving, and updating data in the database

Its main component are:

- **Authorization & integrity manager**: authority of users, integrity constraints
- **File manager**: Allocation of disk space & data structures.
- **Buffer manager**: Disks to main memory fetching, what data to cache.

Storage manager

Several data structures

- **Data files**: To store the database
- **Data dictionary**: To store meta data
- **Index**: Provide fast access to data items

The Query Processor

- **DDL interpreter**: Interprets DDL statements and records the definitions in the data dictionary.
- **DML compiler**: Translates a DML statements in a query language into an evaluation plan consisting of low-level instructions [Query optimization]
- **Query evaluation engine**: Executes low-level instructions generated by the DML compiler

Transaction manager

A transaction is a collection of operations that performs a single logical function in a database application

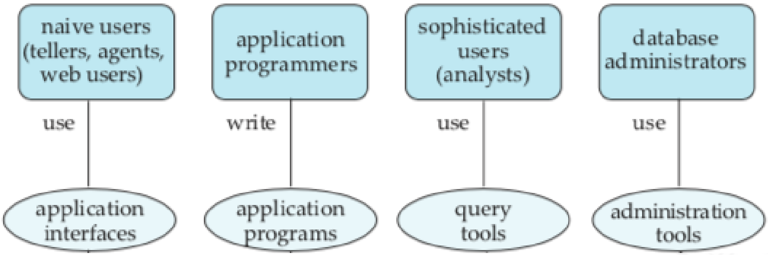
- Each transaction is a unit of both atomicity and consistency
- During the execution of a transaction, it may temporarily allow inconsistency **Any Example?**

Transaction manager

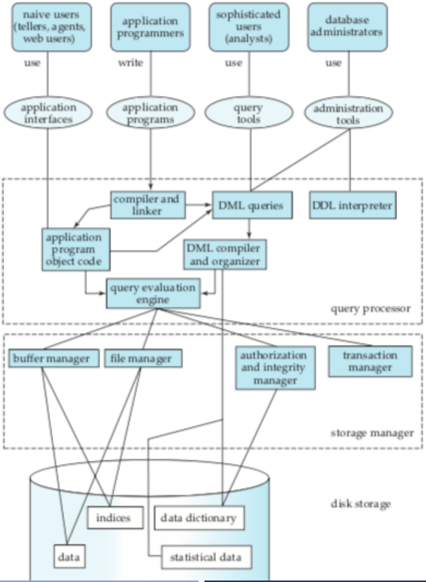
It has two components:

- **Transaction-management**: Database remains in a consistent (correct) state despite system failures
- **Concurrency-control**: It controls the interaction among the concurrent transactions.

Database Users



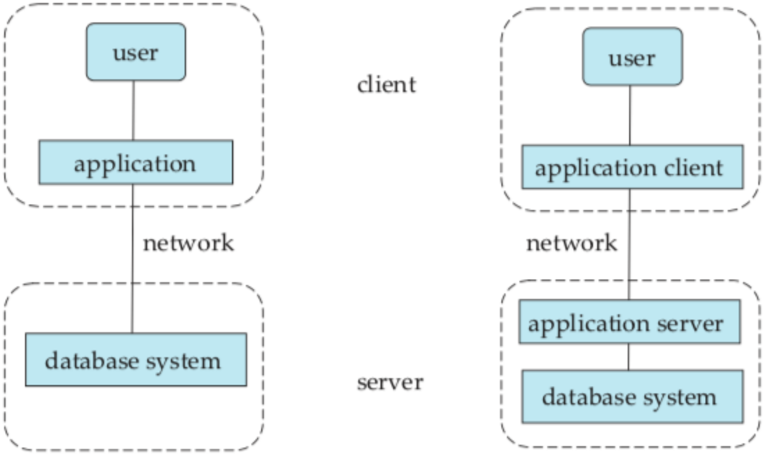
System structure



Database Architecture

- Centralized
- Client-server
- Parallel (multi-processor)
- Distributed

Client-server Architecture



(a) Two-tier architecture

(b) Three-tier architecture

The ACID properties

- **Atomicity**: Either succeeds completely, or fails completely.
- **Consistency**: Any data written to the database must be valid according to all defined rules, including constraints, cascades, triggers, and any combination thereof.
- **Isolation**: Concurrent execution of transactions leaves the database in the same state that would have been obtained if the transactions were executed sequentially.
- **Durability**: Guarantees that once a transaction has been committed, it will remain committed even in the case of a system failure.

Acknowledgments/Contributions

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